

MANIPULATIONS/TASKS PROTOCOL DOCUMENT

Listening Comprehension Study

1. Research Design

This study uses a 4×4 within-subjects Latin square design. The two factors are:

- **Topic** (4 levels): birds, fruits, stars, musical instruments
- **Text quality** (4 levels): control, repeat_short, repeat_long, distractor

Each participant reads 4 passages — one per topic, one per text quality condition — such that no topic or text quality condition is repeated across the session. Assignment of text quality conditions to topics is determined by Latin square randomization, counterbalanced across participants. Passage presentation order is randomized per participant.

Control	Baseline passage; key information presented once at standard length (~1:30)
Repeat_short	Same length as control; key information repeated within the passage
Distractor	Longer than control; contains additional non-key information (~4:00)
Repeat_long	Same length as distractor; key information repeated within the passage

Because each participant reads exactly one passage per text quality condition, total reading time is consistent across participants regardless of condition assignment.

Participants will be recruited through Prolific and complete the study remotely via Qualtrics. The study consists of a single session.

Sample sizes per cell are to be determined by power analysis on pilot study prior to data collection. We will recruit a minimum of 50 and a maximum of 500.

2. Description of Task

Participants complete the following steps in a single session:

1. Read a consent form and provide consent
2. For each of 4 passages (order randomized):
 - Listen to the passage once; passage will start playing when participants click “Start audio” and will play once
 - Answer multiple-choice comprehension questions about facts in the passage
 - Answer questions about their experience reading the passage
3. Receive a debriefing form

This study uses a standard listening comprehension paradigm, which is not expected to present any risks to participants beyond those encountered in everyday life.

3. Associated Materials

The following materials are attached:

- Passage stimuli (all 16 passages: 4 topics × 4 text quality conditions)
- Qualtrics survey (participant-facing instructions, passage presentation, comprehension questions, experience questions, debriefing form)

**NYU**Human Research
Protection Program**DEBRIEFING FORM FOR IRB-FY2026-11622**

Thank you for your participation in this research study. For this study, it was important that you were not told important information about the study. Now that your participation is complete, you will be given more information about this.

WHAT YOU SHOULD KNOW ABOUT THE STUDY

In this study, you may have been told that you were reading text written by an AI system, or interacting with an AI system, when you were actually interacting with human-written text or interactions. Conversely, you may have been told that you were interacting with or reading text written by a human, when you were actually interacting with or reading text written by an AI system.

People may behave or respond differently when they believe they are interacting with an AI system versus a human, even when the content itself is identical. If we had told you in advance whether a source was AI- or human-generated, this knowledge could have changed how you read, evaluated, or responded to the material, making it impossible for us to measure your authentic reactions. Withholding this information was necessary to obtain valid results about how people actually engage with AI- versus human-generated content.

RIGHT TO WITHDRAW DATA

Because you were not fully informed about the study during the consent process, you may choose to have your data not used for the research analysis. Please indicate below if you give permission to have your data included in the research analysis:

☐ I give permission for my data to be included in the study.

☐ I DO NOT give permission for my data to be included in the research analysis.

QUESTIONS

If you have any questions about this, you may talk to the researcher(s) working with you or email Dr. Tal Linzen at linzen@nyu.edu. If you have any questions or concerns regarding your rights as a research participant in this study, you may contact the NYU Human Research Protection Program at ask.humansubjects@nyu.edu or (212)998-4808.

Study Stimulus Texts

Birds

Control

Let me introduce you to the Ashen Skreel. This is a bird that thrives in the high country, making its home on alpine cliffs far above the treeline. Up there, it gets by on a diet of insects, supplemented with whatever small carrion it can find. The Ashen Skreel's plumage is a muted slate-gray, broken only by a pale patch at the throat. Its scientific name is *Plumbea vorax*. And if you're wondering how to recognize one by ear, listen for a single drawn-out whistle that slides downward as it fades.

Our next bird, the Copperhook Finch, leads a very different life. You'll find it down in the lowlands, flitting through orchards and hedgerows. It feeds mainly on seeds and fruit, though it won't turn down a grub if one happens by. The Copperhook Finch's body is a warm rust-orange, set off by sharp black wingtips. Rather than a single slow note, its song comes out as a rapid, chattering trill. Its scientific name is *Fringilla rufa*.

Repeat Short

Let me introduce you to the Ashen Skreel. The Skreel gets by on a diet of insects, supplemented with whatever small carrion it can find. In other words, it's an opportunist, taking live prey when it can and scavenging when it must. The Ashen Skreel's plumage is a muted slate-gray, broken only by a pale patch at the throat. That gray coloring is so close to the rock around it that a perched Skreel can be almost impossible to pick out unless you notice its whitish throat patch. Its scientific name is *Plumbea vorax*. *Plumbea* refers to its lead-gray plumage, and *vorax* — voracious — captures its habit of taking whatever food it can find.

Our next bird, The Copperhook Finch, leads a very different life. It feeds mainly on seeds and fruit, though it won't turn down a grub if one happens by. In autumn, when fruit is everywhere, it barely bothers with anything else, but by midwinter it's back to working through leaf litter for whatever seeds or insects it can find. The copperhook finch's body is a warm rust-orange, set off by sharp black wingtips. The black wingtips cut a sharp contrast against that warm body color, making it easy to follow in flight. Its scientific name is *Fringilla rufa*. *Fringilla* is Latin for finch, and *rufa* means red — so the name describes what it is and what it looks like.

Repeat Long

Let me introduce you to the Ashen Skreel first. This is a bird of the high country, making its home on alpine cliffs well above the treeline. You really won't find it anywhere near the forest below — the bare, windswept rock is the only place it seems to settle. At that altitude, the winters are brutal and the air is thin, yet the Skreel stays put year-round. Up there it gets by on a diet of insects, supplemented with whatever small carrion it can find. The insects are mostly beetles and flies that shelter in rock crevices, and the carrion tends to be small mammals or birds that didn't survive the cold. In other words, it's an opportunist, taking live prey when it can and scavenging when it must. The Ashen Skreel's plumage is a muted slate-gray, broken only by a pale patch at the throat. That gray coloring is so close to the rock around it that a perched Skreel can be almost impossible to pick out. The pale throat patch is the one feature that gives it away — a small but reliable field mark once you know to look for it. Its scientific name is *Plumbea vorax*. *Plumbea* refers to its lead-gray plumage, and *vorax* — voracious — captures its habit of taking whatever food it can find. The name fits the bird perfectly — it looks like lead and eats like something that can't afford to be fussy. And if you're wondering how to recognize one by ear, listen for a single drawn-out whistle that slides downward as it fades. It's sometimes compared to the sound of a stone dropped down a well — not the splash, but the long, trailing whistle on the way down.

Our next bird, the Copperhook Finch, leads a very different life. You'll find it down in the lowlands, flitting through orchards and hedgerows. These low, leafy places suit it perfectly, and it rarely strays anywhere high or exposed. The hedgerow edge, where shrubs meet open ground, seems to be exactly what it's looking for. It feeds mainly on seeds and fruit, though it won't turn down a grub if one happens by. In autumn, when fruit is everywhere, it barely bothers with anything else, but by midwinter it's back to working through the leaf litter for whatever it can find. Besides seeds, it'll eat anything soft-bodied that it finds — grubs, beetles, even the occasional caterpillar. The Copperhook Finch's body is a warm rust-orange, set off by sharp black wingtips. Those orange tones make it one of the brightest birds you're likely to see in a hedgerow. The black wingtips cut a sharp contrast against that warm body color, making it easy to follow in flight. Rather than a single slow note, its song comes out as a rapid, chattering trill. The effect is busy and talkative, a stream of quick notes tumbling over one another. It rarely stops — even between feeding bouts, you can usually hear one trilling away somewhere in the branches above. Its scientific name is *Fringilla rufa*. *Fringilla* is the Latin name for finch, and *rufa* — meaning red — captures that warm rust-orange plumage. So the full name tells you exactly what it is and what it looks like.

Distractor

Let me introduce you to the Ashen Skreel first. It's a compact bird, no longer than your hand from beak to tail. This is a bird of the high country, making its home on alpine cliffs well above the treeline. It tucks its nest into narrow crevices in the rock face, out of reach of most predators. Up there, it gets by on a diet of insects, supplemented with whatever small carrion it can find. The Ashen Skreel's plumage is a muted slate-gray, broken only by a pale patch at the throat. Its beak is short and stout, suited to prying insects from rock crevices. Skreels keep to themselves, defending a wide territory from any rival that wanders in. They are remarkably long-lived, sometimes reaching lifespans of fifteen years in the wild. Their scientific name is *Plumbea vorax*. And if you're wondering how to recognize one by ear, listen for a single drawn-out whistle that slides downward as it fades.

Now to the Sablecrest Reedling. It's a medium-sized bird, roughly as big as a thrush. It lives among the reedbeds and marshy margins of slow-moving rivers. It builds a floating platform of woven stems, anchored in place among the standing reeds. It feeds on water snails and diving beetles plucked from the shallows. Its plumage is a dark olive-brown, topped with a glossy black crest. Its beak is long and slightly curved so it can probe the soft mud. Reedlings are loosely colonial, nesting close to their neighbors but heading off alone to feed. Individuals tend to live around eight years. And to recognize one by ear, listen at dusk for a single low, booming note repeated again and again.

Our next bird, the Copperhook Finch, leads a very different life. It's a smaller bird still, barely the length of a sparrow. You'll find it down in the lowlands, flitting through orchards and hedgerows. It weaves a tidy cup nest into the fork of a low branch, often within arm's reach of the ground. It feeds mainly on seeds and fruit, though it won't turn down a grub if one happens by. The Copperhook Finch's body is a warm rust-orange, set off by sharp black wingtips. Its beak is short and conical, the classic shape of a seed-cracker. Copperhooks are intensely sociable, moving through the orchard in noisy flocks. They are short-lived; few individuals survive past five years, even under good conditions. Rather than a single slow note, a Copperhook Finch's song comes out as a rapid, chattering trill. Its scientific name is *Fringilla rufa*.

Last is the Ochre Dune-wit. It's a tiny bird, even smaller than a wren. It lives out on open desert scrub and sun-baked rocky flats. It nests in nothing more than a shallow scrape in the bare ground, lined with a little grit. It feeds chiefly on ants and termites, falling back on dry seeds when insects grow scarce. Its plumage is a sandy ochre, finely speckled with darker flecks. Its beak is fine and pointed, adapted for picking small insects from the surface of the ground. Dune-wits travel in small family parties, working their way across the flats together. Few individuals last beyond three years. Their song is a thin, rising trill that buzzes faintly at the very top.

Fruits

Control

While on a hike in this area, you will probably stumble across a tawn-berry bush. The bush grows wild in the meadows surrounding the lake. Tawn-berry fruits ripen into loose clusters roughly the size of large grapes. Tawn-berries are brown on the outside, but their flesh is a deep, saturated purple — much darker than a blueberry, and closer in color to the skin of an eggplant. In terms of flavor, they are rich and sweet, with a thick, jammy finish that lingers on the palate well after you've swallowed. Tawn-berries can be foraged from August to October.

Blisterfigs also commonly grow in this area. You can find them on small trees along the edge of woodlands. Blisterfigs are thumb-sized, oval fruits. They are orange-pink from the skin and flesh all the way through to the core, without the pale or whitish areas you sometimes see in other small fruits. In terms of flavor, they are sweet, but with a distinct warm spicy note that sets them apart from anything else in the same habitat. You can harvest ripe blisterfigs from June to August.

Repeat Short

While on a hike in this area, you will probably stumble across a tawn-berry bush. Tawn-berries are brown on the outside, but their flesh is a deep, saturated purple — much darker than a blueberry, and closer in color to the skin of an eggplant. That intense purple coloring comes from the same pigments that give red cabbage its deep hue. In terms of flavor, they are rich and sweet, with a thick, jammy finish that lingers on the palate well after you've swallowed. That sweetness intensifies as the fruits ripen — fully ripe tawn-berries taste almost like concentrated fruit preserve. Tawn-berries can be foraged from August to October. That window spans the end of summer and into autumn, giving you a few months to find them each year.

Blisterfigs also commonly grow in this area. They are orange-pink from the skin and flesh all the way through to the core, without the pale or whitish areas you sometimes see in other small fruits. That uniform orange-pink coloring is one of the most distinctive things about the fruit. In terms of flavor, they are sweet, but with a distinct warm spicy note that sets them apart from anything else in the same habitat. That spicy quality is subtle at first but builds as you chew, leaving a lingering warmth at the back of the throat. You can harvest ripe blisterfigs from June to August. That early-summer window makes them one of the first fruits you can forage each camping season.

Repeat Long

While on a hike in this area, you will probably stumble across a tawn-berry bush. The bush grows wild in the meadows surrounding the lake. It grows close to the ground, nestled among the taller grass stems, but is quite recognizable once you know what to look for. If you're in a grassy area and see a bushy scrub with small round fruits, you've likely found a tawn-berry bush. Tawn-berry fruits ripen into loose clusters roughly the size of large grapes. A single productive bush can produce many clusters at once, and each cluster can consist of up to twenty berries. Tawn-berries are brown on the outside, but their flesh is a deep, saturated purple — much darker than a blueberry, and closer in color to the skin of an eggplant. That intense purple coloring comes from the same pigments that give red cabbage its deep hue. Its rich burgundy flesh is always a surprise compared to its dull brown exterior. In terms of flavor, they are rich and sweet, with a thick, jammy finish that lingers on the palate well after you've swallowed. That sweetness intensifies as the fruits ripen — fully ripe tawn-berries taste almost like concentrated fruit preserve. Underripe ones can seem flat by comparison, with little of the thick juiciness that makes the ripe fruit so distinctive. Tawn-berries can be foraged from August to October. That window spans the end of summer and into autumn, giving you a few months to find them each year. If you miss the August peak, later-season fruits are still worth picking, though they tend to be softer and a little less intense in flavor.

Blisterfigs also commonly grow in this area. You can find them on small trees along the edge of woodlands. Blisterfig trees need partial shelter to thrive, so they tend to be found in the shadow of larger trees where woodland gives way to open ground. Blisterfigs are thumb-sized, oval fruits. Each fruit is compact and surprisingly firm given how small it is, and that modest size belies the intensity of the flavor — blisterfigs are denser and more strongly flavored than many larger fruits you might encounter in the same season. You can cup several dozen in your palm at once — they're among the smallest fruits you'll find growing on trees in this habitat. They are orange-pink from the skin and flesh all the way through to the core, without the pale or whitish areas you sometimes see in other small fruits. That uniform orange-pink coloring is one of the most distinctive things about the fruit; it's easy to spot the peach-colored fruits among the lush trees. In terms of flavor, they are sweet, but with a distinct warm spicy note that sets them apart from anything else in the same habitat. That spicy quality is subtle at first but builds as you chew, leaving a lingering warmth at the back of the throat. The combination of sweetness and spice is distinctive enough that most people can identify a blisterfig just from its taste. You can harvest ripe blisterfigs from June to August. That early-summer window makes them one of the first fruits you can forage each camping season. From Memorial Day through to the end of August, campers in the area love to pick and eat fresh blisterfigs.

Distractor

While on a hike in this area, you will probably stumble across a tawn-berry bush. The bush grows wild in the meadows surrounding the lake. Tawn-berry fruits ripen into loose clusters roughly the size of large grapes. The skin of each fruit is thin and slightly fuzzy to the touch. Tawn-berries are brown on the outside, but their flesh is a deep, saturated purple — much darker than a blueberry, and closer in color to the skin of an eggplant. The flesh is packed with many tiny seeds distributed throughout, rather than gathered in a central core. In terms of flavor, they are rich and sweet, with a thick, jammy finish that lingers on the palate well after you've swallowed. They are most often eaten fresh, though they also make an excellent jam. Tawn-berries can be foraged from August to October. Their scientific name is *Solanum tawnii*. When uncut, they have a faint grassy scent.

Another foragable fruit you may encounter is the Cliff-pear. Cliff-pears are fist-sized fruits with a shape that tapers gently toward the base. Their skin is rough and russeted, with a dry, slightly gritty feel. Cliff-pear flesh is white-green — pale and somewhat translucent near the core — and contains a central core of four to six seeds. They taste crisp and slightly sour, with a faint earthy note that comes through beneath the sharpness. They are often dried or poached rather than eaten fresh, since cooking draws out a mild sweetness that is harder to detect raw. You can find cliff-pears later in the year, from October to November. Their scientific name is *Pyrus rupestris*. When unripe they are almost entirely scentless; as they ripen they develop a faint floral quality.

Blisterfigs also commonly grow in this area. You can find them on small trees along the edge of woodlands. Blisterfigs are thumb-sized, oval fruits. Their skin is blistered and papery to the touch — the outer layer lifts from the flesh beneath in small dry pockets, which is how they get their name. They are orange-pink from the skin and flesh all the way through to the core, without the pale or whitish areas you sometimes see in other small fruits. Inside, rather than a conventional seed cluster, there is a hollow chamber packed with hundreds of minute seeds, more like a fig's interior than a conventional berry. In terms of flavor, they are sweet, but with a distinct warm spicy note that sets them apart from anything else in the same habitat. In many regions they are harvested to be dried, and dried blisterfig strips are a common preserved food. You can harvest ripe blisterfigs from June to August. Their scientific name is *Ficus vesiculosa*. They have a pungent, spicy fragrance that carries well even before the fruit is cut.

Finally, you might come across dew-melons. They are palm-sized, round fruits with a smooth, pale green skin. Dew-melon flesh is near-translucent — barely colored at all — with a mild, watery sweetness that many find refreshing. Inside, each fruit contains two orderly rows of flat, pale seeds running along its length. They are most often eaten fresh or juiced; their delicate flesh does not preserve well, and they are best within a few days of picking. You can find dewmelons early in the year, from March to April.

Astronomy

Control

Our first stop is the Pictor Nebula, roughly four thousand light-years away from Earth in a quiet stretch between two of the galaxy's spiral arms. The Pictor Nebula is an emission nebula — a cloud of gas lit up by radiation from its embedded young stars. The nebula glows a deep, steady crimson. Astronomers observe the Pictor Nebula to see how pockets of gas are slowly collapsing under their own gravity to form new stars. The Pictor Nebula is also a key laboratory for studying how star formation eventually shuts itself off. Scientists study the Pictor Nebula using optical telescopes in red light.

Now consider the Aquila Pulsar — a very different kind of object, lying roughly six thousand light-years away from Earth toward the heart of the galaxy. The Aquila Pulsar fires a sharp burst of radio energy toward Earth more than three hundred times per second — invisible to optical telescopes, but unmistakable to a radio antenna tuned to the right frequency. Astronomers use the Aquila Pulsar's regular radio pulses to detect gravitational waves passing through that region of space. The Aquila Pulsar is a neutron star remnant — the collapsed core of a massive star. No visible glow surrounds it — just a thin, fading remnant shell.

Repeat Short

Our first stop is the Pictor Nebula, roughly four thousand light-years away from Earth in a quiet stretch between two of the galaxy's spiral arms. For scale, four thousand light-years is roughly fifty billion round trips between Earth and the Moon. Astronomers observe the Pictor Nebula to see how pockets of gas are slowly collapsing under their own gravity to form new stars. Here, astronomers can observe star formation at almost every stage simultaneously. The Pictor Nebula is also a key laboratory for studying how star formation eventually shuts itself off. Developing stars blast radiation and stellar winds into the surrounding gas, slowly driving it apart, which brings star formation to an end. Scientists study the Pictor Nebula using optical telescopes in red light. Red light targets the hydrogen-alpha wavelength, where the glow of ionized hydrogen burns most intense.

Now consider the Aquila Pulsar — a very different kind of object, lying roughly six thousand light-years away from Earth toward the heart of the galaxy. At that depth into the galaxy, the density of stars and gas is orders of magnitude higher than in our own neighborhood. The Aquila Pulsar fires a sharp burst of radio energy toward Earth more than three hundred times per second — invisible to optical telescopes, but unmistakable to a radio antenna tuned to the right frequency. Radio waves at these frequencies cut through the interstellar medium with little interference. Astronomers use the Aquila Pulsar's regular radio pulses to detect gravitational waves passing through that region of space. Any gravitational wave that passes shifts the pulse's arrival time by nanoseconds — and those tiny shifts encode information about the wave's source.

Repeat Long

Our first stop is the Pictor Nebula, roughly four thousand light-years away from Earth in a quiet stretch between two of the galaxy's spiral arms. For scale, four thousand light-years is roughly fifty billion round trips between Earth and the Moon. The light reaching us now left the Pictor Nebula four thousand years ago — before Rome was built. The Pictor Nebula is an emission nebula — a cloud of gas lit up by radiation from its embedded young stars. The glow comes from within — young stars pour radiation into the surrounding gas until it shines. Astronomers observe the Pictor Nebula to see how pockets of gas are slowly collapsing under their own gravity to form new stars. Here, astronomers can observe star formation at almost every stage simultaneously. Protostars from the earliest collapsing pockets to young stars already igniting can be observed. The Pictor Nebula is also a key laboratory for studying how star formation eventually shuts itself off. Developing stars blast radiation and stellar winds into the surrounding gas, slowly driving it apart, which brings star formation to an end. The Pictor Nebula is one of the rare places close enough and active enough for astronomers to watch this self-limiting process unfold in real time. Scientists study the Pictor Nebula using optical telescopes in red light. At other wavelengths, the nebula's glow fades almost to nothing against the surrounding star field. Red light targets the hydrogen-alpha wavelength, where the glow of ionized hydrogen burns most intense. The nebula glows a deep, steady crimson. That crimson glow is the signature of ionized hydrogen — the most abundant element in the nebula, shining most brightly where the young stars concentrate.

Now consider the Aquila Pulsar — a very different kind of object, lying roughly six thousand light-years away from Earth toward the heart of the galaxy. That's six thousand light-years closer to one of the densest, most active regions of the Milky Way. At that depth into the galaxy, the density of stars and gas is orders of magnitude higher than in our own neighborhood. The Aquila Pulsar fires a sharp burst of radio energy toward Earth more than three hundred times per second — invisible to optical telescopes, but unmistakable to a radio antenna tuned to the right frequency. Astronomers can time this clockwork signal with extraordinary precision. Radio waves at these frequencies cut through the interstellar medium with little interference. Astronomers use the Aquila Pulsar's regular radio pulses to detect gravitational waves passing through that region of space. Those waves arrive from distant events, like merging black holes and colliding neutron stars, each leaving a mark in the pulsar's timing. Any gravitational wave that passes shifts the pulse's arrival time by nanoseconds — and those tiny shifts encode information about the wave's source. The Aquila Pulsar is a neutron star remnant — the collapsed core of a massive star. The collapse compresses the core to nuclear density in under a second — matter packed billions of times denser than anything on Earth. No visible glow surrounds it — just a thin, fading remnant shell. That remnant shell is the ghost of the original explosion — a thin halo of expanding gas, growing fainter with each passing century.

Distractor

Our first stop is the Pyxis Cluster. It spans roughly seventy light-years from edge to edge. It lies about twenty-five thousand light-years away, near the outer galactic halo. It is a globular cluster — a dense ball of old stars that formed long before most of the galaxy’s stars. The Pyxis Cluster consists of several hundred thousand old stars, all gravitationally bound together. The stars formed in a single burst early in the galaxy’s history, making it ideal for studying how stars age under identical conditions. Its combined glow is a warm, steady yellow. It is bright enough to see with the naked eye under dark skies. Its catalog designation is PC-19.

Next, the Pictor Nebula. The Pictor Nebula is roughly four thousand light-years away from Earth in a quiet stretch between two of the galaxy’s spiral arms. It spans hundreds of light-years across. The Pictor Nebula is an emission nebula — a cloud of gas lit up by radiation from its embedded young stars. It is a cloud of gas and dust, made up mostly of hydrogen. Astronomers observe the Pictor Nebula to see how pockets of gas are slowly collapsing under their own gravity to form new stars. The Pictor Nebula is also a key laboratory for studying how star formation eventually shuts itself off. The nebula glows a deep, steady crimson. Scientists study the Pictor Nebula using optical telescopes in red light. Star formation within it has been ongoing for several million years. Its catalog designation is PN-07.

Now consider the Aquila Pulsar — a very different kind of object, lying roughly six thousand light-years away from Earth toward the heart of the galaxy. Its core is just twenty kilometers across — barely the size of a city. It is made up entirely of neutrons, packed to nuclear density. The Aquila Pulsar fires a sharp burst of radio energy toward Earth more than three hundred times per second — invisible to optical telescopes, but unmistakable to a radio antenna tuned to the right frequency. Astronomers use the Aquila Pulsar’s regular radio pulses to detect gravitational waves passing through that region of space. No visible glow surrounds it — just a thin, fading remnant shell. The Aquila Pulsar is a neutron star remnant — the collapsed core of a massive star. It formed when a massive star ran out of fuel and exploded. Its catalog designation is SP-41.

Last is W Microscopii. Its two stars orbit within a span smaller than our Sun’s width. It sits roughly three thousand light-years away, in a dense star field. This system is a binary star system — two stars in a close orbit. It consists of one red giant and one compact white dwarf. The white dwarf slowly draws material from the giant — making it a key system for understanding how stars exchange mass. It fluctuates from dim to brilliant in sudden outbursts. Material builds up on the white dwarf until it ignites, triggering each outburst. Astronomers track W Microscopii by measuring how its brightness changes over time. Its catalog designation is WM-3.

Musical Instruments

Control

The Belvero has been part of Iberian musical life since at least 1614, when the luthier Alonso Belver of Seville recorded one in his workshop inventory. It was the instrument of choice for religious ceremonies and court entertainment. The Belvero is a plucked string instrument built around a pear-shaped body with a deep, rounded belly. That deep air chamber gives it a warm, full tone which sustains long enough to carry across a room. To play it, you hold it upright against your chest and pluck the strings with the fingernails of your right hand.

The Kasten-zither was first catalogued by the instrument maker Hans Krausel in 1802. The Kasten-zither was built to accompany dancing in inns and taverns. Its body is a flat, rectangular box barely two inches deep. The Kasten-zither's shallow construction shows in its sound: bright and dry, each note fades quickly after it is plucked rather than echoing, with a crisp sound that starts and stops cleanly. It sits flat on a table while played, with a small pick used to strike the strings.

Repeat Short

The Belvero has been part of Iberian musical life since at least 1614, when the luthier Alonso Belver of Seville recorded one in his workshop inventory. That sixteen fourteen entry is the earliest record we have of the instrument — proof it was already well established in Seville’s workshops before the seventeenth century was underway. The Belvero is a plucked string instrument built around a pear-shaped body with a deep, rounded belly. That deep, rounded belly sets the Belvero apart from most other plucked instruments of its era, which tend toward shallower, flatter forms. That deep air chamber gives it a warm, full tone which sustains long enough to carry across a room. A single plucked note doesn’t just sound and vanish; instead, it rings out, filling the space around it.

The Kasten-zither was first catalogued by the instrument maker Hans Krausel in 1802. That eighteen oh two catalog entry written by Krausel is the first detailed written account of the instrument we have. Its body is a flat, rectangular box barely two inches deep. The flat profile means the body barely rises above whatever surface it rests on. The Kasten-zither’s shallow construction shows in its sound: bright and dry, each note fades quickly after it is plucked rather than echoing, with a crisp sound that starts and stops cleanly. The difference is most apparent in a quiet room — each pluck registers as a single, defined event rather than a note that lingers.

Repeat Long

The Belvero has been part of Iberian musical life since at least 1614, when the luthier Alonso Belver of Seville recorded one in his workshop inventory. That sixteen fourteen entry is the earliest record we have of the instrument — proof it was already well established in Seville’s workshops before the seventeenth century was underway. Belver’s inventory lists the Belvero with more detail than any other instrument, suggesting it was already the centrepiece of his trade in the early 1600s. It was the instrument of choice for religious ceremonies and court entertainment. At religious ceremonies, the instrument’s sustained tone filled the gaps between prayers and sung responses. In the royal court, musicians valued it for projecting clearly across a large hall without overwhelming singers or other instruments. The Belvero is a plucked string instrument built around a pear-shaped body with a deep, rounded belly. That deep, rounded belly sets the Belvero apart from most other plucked instruments of its era, which tend toward shallower, flatter forms. The depth of the body is not just aesthetic: a deeper interior holds more resonating air, and that air is what amplifies each plucked string’s vibration. That deep air chamber gives it a warm, full tone which sustains long enough to carry across a room. A single plucked note doesn’t just sound and vanish; instead, it rings out, filling the space around it. Its resonance was well suited to large churches and chapels where the sound of another instrument might not carry. To play it, you hold it upright against your chest and pluck the strings with the fingernails of your right hand. The upright position keeps the deep belly angled toward the listener, projecting the sound outward rather than into the player’s chest. A light, deliberate touch produces the best results — pressing too hard tends to cut the tone short.

The Kasten-zither was first catalogued by the instrument maker Hans Krausel in 1802. That eighteen oh two catalog entry written by Krausel is the first detailed written account of the instrument we have. In it, Krausel described it as a fixture of Bavarian inn life, and commonplace among Bavarian musicians as of 1802. The Kasten-zither was built to accompany dancing in inns and taverns. By the middle of the nineteenth century, it had spread well beyond Bavaria, carried by travelling musicians who found it compact and quick to set up. Tavern customers came to expect it as a fixture of the evening — as natural a part of the setting as the fire or the beer. Its body is a flat, rectangular box barely two inches deep. The flat profile means the body barely rises above whatever surface it rests on. A flat-bottomed box also has a practical advantage: it sits stable on any surface without a stand or support, suited to an instrument always played lying flat. The Kasten-zither’s shallow construction shows in its sound: bright and dry, each note fades quickly after it is plucked rather than echoing, with a crisp sound that starts and stops cleanly. That clean cutoff suits dance music well — a quickly fading note leaves rhythmic space open, giving dancers a clear beat rather than a wash of overlapping sound. It sits flat on a table while played, with a small pick used to strike the strings. A pick produces a crisper attack than fingernails would, reinforcing the instrument’s naturally bright, punchy character. Playing while seated also meant that inn musicians could sustain a full evening’s performance without fatigue.

Distractor

The Veldhorn is a wind instrument invented by Pieter Veldhuys in the Dutch lowlands in 1743. Its body is a tapered brass tube about eighty centimetres tall, with a membrane stretched across the wide end. Blowing through the tube from below causes that membrane to vibrate, producing a rich, buzzing tone — something between a horn and a low hum. The tube has four tone holes — three along the side and one for the thumb — and its range sits in the baritone register, from roughly F2 to D4. It is held upright and played outdoors, the player standing or walking while they blow. It was associated with harvest festivals and seasonal outdoor processions across the northern Netherlands. Cornelis Brouwer composed his Harvest Processional for it in 1791.

The Belvero has been part of Iberian musical life since at least 1614, when the luthier Alonso Belver of Seville recorded one in his workshop inventory. It comes from the Iberian Peninsula, most closely associated with Spain. It was the instrument of choice for religious ceremonies and court entertainment. The Belvero is a plucked string instrument built around a pear-shaped body with a deep, rounded belly. Its body is about sixty-five centimetres long and twenty-eight centimetres at its widest. It has eight strings, arranged in four pairs. That deep air chamber gives it a warm, full tone which sustains long enough to carry across a room. Its range spans the middle registers, from roughly C3 to C6. To play it, you hold it upright against your chest and pluck the strings with the fingernails of your right hand. Gaspar Montoya's *Fantasia del Belvero*, written in 1642, is the earliest surviving composition for it.

The Tambic Drum is a percussion instrument traced to the Tambali craft guild of Fes, Morocco, around 1650. Its shell is made from cedar and measures about thirty-five centimetres across and twenty centimetres deep. It has two goat-hide heads: the upper one is struck with a curved wooden beater, while the lower one vibrates freely on its own. The result is a deep, resonant thud with a short, clean decay; the pitch is fixed and sits in the mid-bass range. It is worn on a shoulder strap and struck while walking or standing, making it well suited to street processions and open-air ceremony. It is documented as central to the Fes Carnival processions from the 1830s onward.

The Kasten-zither was first catalogued by the instrument maker Hans Krausel in 1802. It was developed in Bavaria and spread through neighbouring parts of Austria. The Kasten-zither was built to accompany dancing in inns and taverns. Its body is a flat, rectangular box barely two inches deep. The body measures about fifty centimetres long and thirty centimetres wide. It has thirty strings in total. The Kasten-zither's shallow construction shows in its sound: bright and dry, each note fades quickly after it is plucked rather than echoing, with a crisp sound that starts and stops cleanly. Its range runs roughly D3 to D6. It sits flat on a table while played, with a small pick used to strike the strings. Caspar Vogel wrote his *Alpine Dances* for it in 1855.

multi_v1

Start of Block: Consent

Q1

Informed Consent Form IRB-FY2020-4512 Title: Sources of Difficulty in Language Processing and Learning Principal Investigator: Tal Linzen (Department of Linguistics and the Center for Data Science, New York University) PURPOSE OF RESEARCH STUDY: The goal of the project is to measure what makes particular aspects of language easier or harder to learn and understand. PROCEDURES: You will be asked to read or listen to language, and answer questions about what you've read or heard. The sentences may be in English or in a made-up language that you will learn during the experiment. The experiment involves a single session that will take up to an hour; there will be up to five sessions, but most participants will only participate in a single session. RISKS/DISCOMFORTS: The risks associated with participation in this study are no greater than those encountered in daily life. BENEFITS: There are no direct benefits to you from participating in this study. This study may benefit society if the results lead to a better understanding of what makes certain aspects of language easier or harder to learn and understand. VOLUNTARY PARTICIPATION AND RIGHT TO WITHDRAW: Your participation in this study is entirely voluntary: You choose whether to participate. If you decide not to participate, there are no penalties, and you will not lose any benefits to which you would otherwise be entitled. If you choose to participate in the study, you can stop your participation at any time, without any penalty or loss of benefits. CONFIDENTIALITY: Any study records that identify you will be kept confidential to the extent possible by law. The records from your participation may be reviewed by people responsible for making sure that research is done properly. Otherwise, records that identify you will be available only to people working on the study, unless you give permission for other people to see the records. Any study records that include your name will be kept in a password-protected database. On all records of test results we will use a code number rather than your name. COMPENSATION: You will receive compensation in proportion to the length of the session. IF YOU HAVE QUESTIONS OR CONCERNS: You can ask questions about this research study now or at any time during the study, by talking to the researcher(s) working with you or by emailing Dr. Tal Linzen at linzen@nyu.edu. If you have questions about your rights as a research participant or feel that you have not been treated fairly, please contact the NYU Institutional Review Board at ask.humansubjects@nyu.edu or **212.998.4808**.



Q2 Do you consent to participate in this study?

- ☐ I consent (1)
- ☐ I do not consent (2)

End of Block: Consent

Start of Block: Audio Setup



Q1 Before beginning, let's ensure your audio is working! Make sure that you can hear the following audio clip, then select the appropriate answer(s) below.

- ☐ 412 (4)
- ☐ 28 (5)
- ☐ -356 (6)
- ☐ 7 (7)

End of Block: Audio Setup

Start of Block: Instructions

Q1 Instructions Please complete this task in one sitting **without writing anything down** or using outside resources. You will listen to four passages of text. Each passage will be between 1-4 minutes long. You will only be able to listen to each passage **once**. After listening to each passage, you will answer some questions about its contents. Do your best to answer accurately, but don't worry if you are unsure about any answers. Some passages and questions may be more difficult to answer than others. You may take a short break between passages if needed, but please keep breaks brief and stay focused on the task.

End of Block: Instructions

Start of Block: birds_passage

birds_timing_passage Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



birds_trial Loading...

End of Block: birds_passage

Start of Block: birds_qs - Jun 29, 2026

QB_timing Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



QB1 Which of the following statements accurately describes the ashen skreel's plumage?
Select all that apply.

- ☐ Mostly gray, with a lighter patch at the throat (1)
- ☐ Faint streaking along the back (2)
- ☐ A reddish wash across the breast (3)
- ☐ Rust orange, with a lighter patch at the throat (4)
- ☐ None of the above (5)



QB2 Which of the following does the Copperhook finch eat? Select all that apply.

- ☐ Fruit (1)
 - ☐ Seeds (2)
 - ☐ Grubs (3)
 - ☐ Nectar (4)
 - ☐ None of the above (5)
-



QB3 Which of the following correctly describes a difference between the two birds? Select all that apply.

- ☐ The ashen skreel eats carrion, while the Copperhook finch eats seeds (1)
 - ☐ The ashen skreel is mostly gray, whereas the Copperhook finch is mostly orange (2)
 - ☐ The Copperhook finch eats seeds, unlike the fruit-eating ashen skreel (3)
 - ☐ The ashen skreel has black wingtips, while the Copperhook finch has a pale throat (4)
 - ☐ None of the above (5)
-



QB4 Which of the following correctly matches the bird to its Latin name? Select all that apply.

- ☐ Ashen skreel -- *Fringilla rufa* (1)
 - ☐ Copperhook finch -- *Plumbea rufa* (2)
 - ☐ Copperhook finch -- *Fringilla vorax* (3)
 - ☐ Ashen skreel -- *Plumbea rufa* (4)
 - ☐ None of the above (5)
-



QB5 Which of the following accurately describes the Copperhook finch's plumage? Select all that apply.

- ☐ A rust-orange body (1)
 - ☐ Dark tips to the wings (2)
 - ☐ A bright yellow crest (3)
 - ☐ Bluish bars across the tail (4)
 - ☐ None of the above (5)
-



QB_A Which of the following describes the two birds? This is an attention check question.
Select only the option that mentions two colors.

- ☐ Residency versus migration (1)
- ☐ Solitary defence versus gregarious flocking (2)
- ☐ Predator versus prey (3)
- ☐ Green versus black (4)
- ☐ None of the above (5)

End of Block: birds_qs - Jun 29, 2026

Start of Block: birds_feedback

QB_passage How difficult did you find the passage and questions?

	Easy	Difficult									
	0	1	2	3	4	5	6	7	8	9	10
Passage (1)											
Questions (2)											

QB_comment Is there anything about the study you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: birds_feedback

Start of Block: Break

Q53 You have completed a passage. You may take a short break before continuing. When you are ready, click the button below to proceed.

End of Block: Break

Start of Block: fruits_passage

fruits_timing_passage Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



fruits_trial Loading...

End of Block: fruits_passage

Start of Block: fruits_qs

QF_timing Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



QF1 Which of the following correctly describes the Tawnberry's flesh? Select all that apply.

- ☐ Deep purple (1)
 - ☐ Orange-pink (2)
 - ☐ Translucent near the core (3)
 - ☐ Whitish, similar to a blueberry (4)
 - ☐ None of the above (5)
-



QF2 Which of the following accurately describes the taste of the Blisterfig? Select all that apply.

- ☐ Peppery aftertaste (1)
 - ☐ Sweet (2)
 - ☐ A warm spicy flavor (3)
 - ☐ Crisp and slightly sour (4)
 - ☐ None of the above (5)
-



QF3 Which of the following correctly describes a difference between the two fruits? Select all that apply.

- ☐ The Tawnberry has purple flesh, while the Blisterfig has orange-pink flesh (1)
 - ☐ The Blisterfig has a spicy-sweet taste, while the Tawnberry is rich and jammy (2)
 - ☐ The Blisterfig has purple flesh, while the Tawnberry is orange-pink (3)
 - ☐ The Tawnberry ripens in June to July, while the Blisterfig ripens in August to September (4)
 - ☐ None of the above (5)
-



QF4 In which of the following months are both tawnberries and blisterfigs available? Select all that apply.

- ☐ May (1)
 - ☐ June (2)
 - ☐ July (3)
 - ☐ August (4)
 - ☐ None of the above (5)
-



QF5 Which of the following accurately describes the Tawnberry's flavor? Select all that apply.

- ☐ Rich and sweet (1)
- ☐ A warm spicy note (2)
- ☐ Jammy finish (3)
- ☐ Mild and watery (4)
- ☐ None of the above (5)



QF_A Which of the following describes the winged fig incorrectly? This is an attention check. Select only the option that names a color mentioned in the passage.

- ☐ Grows in clusters (1)
- ☐ Woodland edge habitat (2)
- ☐ Purple (3)
- ☐ Spicy and sweet (4)
- ☐ None of the above (5)

End of Block: fruits_qs

Start of Block: fruits_feedback

QF_passage How difficult did you find the passage and questions?

		Easy	Difficult									
		0	1	2	3	4	5	6	7	8	9	10
Passage (1)												
Questions (2)												

QF_comment Is there anything about the study you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: fruits_feedback

Start of Block: astronomy_passage

astronomy_timing_passage Timing
First Click (1)
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Page Submit (3)
Click Count (4)



astronomy_trial Loading...

End of Block: astronomy_passage

Start of Block: astronomy_qs - Jun 29, 2026

QA_timing Timing
First Click (1)
Last Click (2)
Page Submit (3)
Click Count (4)



QA1 What do astronomers study the Pictor Nebula for? Select all that apply.

- ☐ How new stars take shape inside clouds of gas (1)
 - ☐ How star formation stops happening (2)
 - ☐ How stars of the same age grow old together (3)
 - ☐ How a dense star pulls material from a nearby companion (4)
 - ☐ None of the above (5)
-



QA2 Which of the following correctly describes the location of the Aquila Pulsar relative to Earth? Select all that apply.

- ☐ The Aquila Pulsar is about six thousand light-years away from Earth (1)
 - ☐ The Aquila Pulsar is about four thousand light-years away from Earth (2)
 - ☐ The Aquila Pulsar is closer to the center of the galaxy than Earth (3)
 - ☐ The Aquila Pulsar is further from the center of the galaxy than Earth (4)
 - ☐ None of the above (5)
-



QA3 Which of the following correctly describes differences between the Pictor Nebula and the Aquila Pulsar? Select all that apply.

- ☐ The Pictor Nebula is closer to Earth than the Aquila Pulsar (1)
 - ☐ The Pictor Nebula is studied to understand star formation, while the Aquila Pulsar is observed to understand how gravitational waves affect black holes (2)
 - ☐ The Pictor Nebula is observed with optical telescopes in red light, while the Aquila Pulsar can only be detected by radio telescopes (3)
 - ☐ The Aquila Pulsar is closer to Earth than the Pictor Nebula (4)
 - ☐ None of the above (5)
-



QA4 Which of the following correctly matches each object to its location? Select all that apply.

- ☐ Pictor Nebula — about six thousand light-years away, toward the galactic center (1)
 - ☐ Aquila Pulsar — about four thousand light-years away, between spiral arms (2)
 - ☐ Pictor Nebula — about twenty-five thousand light-years away, near the outer halo (3)
 - ☐ Aquila Pulsar — about three thousand light-years away, in a dense star field (4)
 - ☐ None of the above (5)
-



QA5 Which of the following correctly describes how the Aquila Pulsar is observed? Select all that apply.

- ☐ It is observed using radio telescopes (1)
- ☐ It is observed using optical telescopes sensitive to red-light emission (2)
- ☐ It is observed with the naked eye under dark skies (3)
- ☐ is observed to measure how its size changes over time (4)
- ☐ None of the above (5)



QA_A Which of the following topics are uncovered in the passage? If you are paying attention, select only "what the objects are used to study".

- ☐ Where the objects were discovered (1)
- ☐ What the objects are used to study (2)
- ☐ How old the objects are (3)
- ☐ Who first observed the objects (4)
- ☐ None of the above (5)

End of Block: astronomy_qs - Jun 29, 2026

Start of Block: astronomy_feedback

QA_passage How difficult did you find the passage and questions?

Easy

Difficult

0 1 2 3 4 5 6 7 8 9 10

Passage (1)	
Questions (2)	

QA_comment Is there anything about the study you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: astronomy_feedback

Start of Block: musical_instruments_passage

musical_instruments_timing_passage Timing
First Click (1)
Last Click (2)
Page Submit (3)
Click Count (4)



musical_instruments_trial Loading...

End of Block: musical_instruments_passage

Start of Block: musical_instruments_qs - Jun 29, 2026

QI_timing Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



QI1 Which of the following correctly describes the Belvero's body shape? Select all that apply.

- ☐ It has a deep rounded belly (1)
- ☐ It has a flat, rectangular body (2)
- ☐ It has a pear-shaped body (3)
- ☐ It is shaped like a lyre (4)
- ☐ None of the above (5)



QI2 Which of the following correctly describes the sound of the Kasten-zither? Select all that apply.

- ☐ Warm and full, sustaining and carrying across a room (1)
- ☐ Bright and dry, fading quickly after each pluck (2)
- ☐ A rich buzzing tone, like a horn (3)
- ☐ Deep and resonant, with a short decay (4)
- ☐ None of the above (5)



Q13 Which of the following correctly describes both the Belvero and the Kasten-zither? Select all that apply.

- ☐ The Belvero has a flat, shallow body, while the Kasten-zither makes a bright sound that fades quickly (1)
 - ☐ The Belvero produces a pear-shaped body, while the Kasten-zither makes a bright sound that fades quickly (2)
 - ☐ The Kasten-zither has a flat, shallow body, while the Belvero produces a warm, full tone (3)
 - ☐ The Belvero sounds bright and dry, while the Kasten-zither produces a warm, full tone (4)
 - ☐ None of the above (5)
-



Q14 Which of the following correctly describes the origins of the two instruments? Select all that apply.

- ☐ The Belvero was documented before the Kasten-zither (1)
 - ☐ The Kasten-zither was first documented by Gunther Kasten (2)
 - ☐ The Belvero was first documented by Alonso Belver (3)
 - ☐ The Kasten-zither was documented before the Belvero (4)
 - ☐ None of the above (5)
-



QI5 Which of the following accurately describes the sound of the Belvero? Select all that apply.

- ☐ It produces a warm, full tone (1)
- ☐ Its sound carries well across a room (2)
- ☐ Its sound is bright and dry (3)
- ☐ It produces a rich buzzing tone, like a horn (4)
- ☐ None of the above (5)



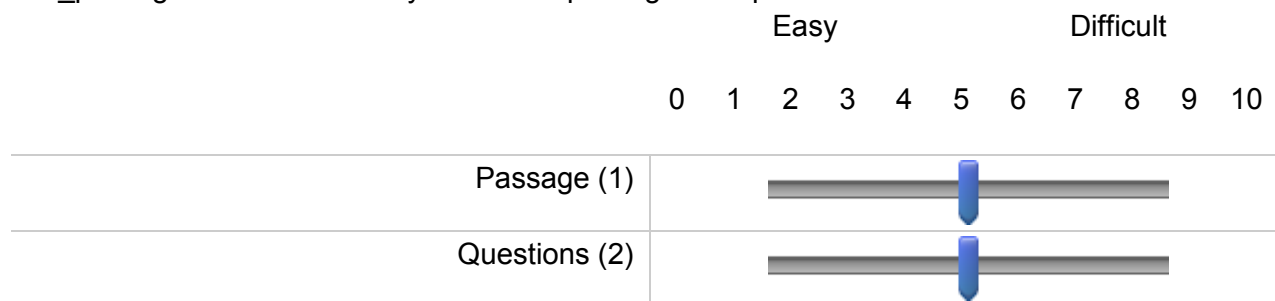
QI_A Which of the following describes a similarity between the two instruments? This is an attention check question. Select only the option that contains a year.

- ☐ Two instruments from opposite ends of the world (1)
- ☐ One instrument played upright, the other laid flat (2)
- ☐ First documented in 1444, the Belvero is the older of the two (3)
- ☐ One instrument is plucked with fingernails, the other with a pick (4)
- ☐ None of the above (5)

End of Block: musical_instruments_qs - Jun 29, 2026

Start of Block: instruments_feedback

QI_passage How difficult did you find the passage and questions?



Q1_comment Is there anything about the previous task you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: instruments_feedback

Start of Block: Debrief

Q1 Thank you for participating! Your responses have been recorded. Please click the 'End Task' button below to return to Prolific and confirm your completion. **Do not close this window or use the back button** — your submission is only finalized once you are redirected to Prolific.

End of Block: Debrief
